

3(a)

A = event that an aircraft is present

$$P(A) = 0.07 \quad P(A^c) = 0.93$$

D = event that aircraft is detected by radar system

$$P(D|A) = 0.98 \quad P(D|A^c) = 0.05$$

$$(i) P(A^c|D) = \frac{P(D|A^c)P(A^c)}{P(D|A^c)P(A^c) + P(D|A)P(A)}$$

$$= \frac{0.05 \times 0.93}{0.05 \times 0.93 + 0.98 \times 0.07}$$

$$= 0.4040$$

A 7%	A ^c 93%
98% D	5% D

$$(ii) P(A|D) = \frac{P(D|A)P(A)}{P(D|A)P(A) + P(D|A^c)P(A^c)}$$

$$= \frac{0.98 \times 0.07}{0.98 \times 0.07 + 0.05 \times 0.93}$$

$$= 0.5960$$

$$(iii) P(D^c|A) = 1 - 0.98 = 0.02$$

$$P(D^c|A^c) = 1 - 0.05 = 0.95$$

A 7%	A ^c 93%
2% D ^c	95% D ^c

$$P(A|D^c) = \frac{P(D^c|A)P(A)}{P(D^c|A)P(A) + P(D^c|A^c)P(A^c)}$$

$$= \frac{0.02 \times 0.07}{0.02 \times 0.07 + 0.95 \times 0.93} = 0.0016$$

$$(iv) P(A^c|D^c) = \frac{P(D^c|A^c)P(A^c)}{P(D^c|A^c)P(A^c) + P(D^c|A)P(A)}$$

$$= \frac{0.95 \times 0.93}{0.95 \times 0.93 + 0.02 \times 0.07} = 0.9984$$

3(b)

$$f(x,y) = \begin{cases} \frac{x^3 y^2}{16}, & 0 \leq x \leq 2, 0 \leq y \leq 2 \\ 0, & \text{otherwise} \end{cases}$$

$$(i) f_x(x) = \int_0^2 \frac{x^3 y^2}{16} dy = \frac{x^3}{16} \left[\frac{y^3}{3} \right]_0^2$$

$$= \frac{x^3}{16} \times \frac{8}{3}$$

$$= \frac{x^3}{6}$$

$$f_x(x) = \begin{cases} \frac{x^3}{6}, & 0 \leq x \leq 2 \\ 0, & \text{otherwise} \end{cases}$$

$$(ii) f_y(y) = \int_0^2 \frac{x^3 y^2}{16} dx = \frac{y^2}{16} \left[\frac{x^4}{4} \right]_0^2$$

$$= \frac{y^2}{16} \frac{16}{4}$$

$$= \frac{y^2}{4}$$

$$f_y(y) = \begin{cases} \frac{y^2}{4}, & 0 \leq y \leq 2 \\ 0, & \text{otherwise} \end{cases}$$

$$(iii) F_x(x) = P(X \leq x) = \int_0^x f_x(x) dx = \frac{1}{6} \left[\frac{x^4}{4} \right]_0^x = \frac{1}{24} x^4$$

$$F_x(x) = \begin{cases} 0 & x < 0 \\ \frac{x^4}{24}, & 0 \leq x < 2 \\ 1, & x \geq 2 \end{cases}$$

(This will not be 1)

$$(iv) F_y(y) = P(Y \leq y) = \int_0^y f_y(y) dy = \int_0^y \frac{y^2}{4} dy = \frac{y^3}{12}$$

$$F_y(y) = \begin{cases} 0, & y < 0 \\ \frac{y^3}{12}, & 0 \leq y < 2 \\ 1, & y \geq 2 \end{cases}$$

(This will not be 1)

Correct

$$f(x, y) = \begin{cases} \frac{x^3 y^3}{16}, & 0 \leq x \leq 2, 0 \leq y \leq 2 \\ 0, & \text{otherwise} \end{cases}$$

$$(i) f_x(x) = \int_0^2 \frac{x^3 y^3}{16} dy = \frac{x^3}{16} \left[\frac{y^4}{4} \right]_0^2 = \frac{x^3}{16} \cdot \frac{2^4}{4} = \frac{x^3}{4}$$

$$f_x(x) = \begin{cases} \frac{x^3}{4}, & 0 \leq x \leq 2 \\ 0, & \text{otherwise} \end{cases}$$

$$(ii) f_y(y) = \int_0^2 \frac{x^3 y^3}{16} dx = \frac{y^3}{16} \left[\frac{x^4}{4} \right]_0^2 = \frac{y^3}{4}$$

$$f_y(y) = \begin{cases} \frac{y^3}{4}, & 0 \leq y \leq 2 \\ 0, & \text{otherwise} \end{cases}$$

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$$F_y(y) = \begin{cases} 0, & y < 0 \\ \frac{y^4}{16}, & 0 \leq y < 2 \\ 1, & y \geq 2 \end{cases}$$